Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18MEDE744]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

FOUNDRY TECHNOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Define casting. Write the classification of casting.	**06**
	b.	Write the steps involved in making a casting.	**08**
2.	**a.**	Explain progressive and directional solidification process	**06**
	b.	Illustrate solidification process of pure metal and alloy with neat sketch.	**08**
3.	**a.**	With a neat sketch explain gating elements.	**08**
	b.	Differentiate between open type riser and blind riser.	**06**
		OR	
4.	**a.**	Explain the factors which controls gating system.	**06**
	b.	List the types of gates and explain any one with neat sketch.	**08**
5.	**a.**	Explain high pressure casting with neat sketch and mentions its advantages and limitation.	**10**
	b.	Write the principles of no bake systems.	**04**
		OR	
6.	**a.**	Explain with a neat sketch explain cupola melting zones and write its applications.	**10**
	b.	Write a note on cokeless cupola.	**04**
7.	**a.**	Explain the casting requirements for aluminium alloys.	**07**
	b.	Describe the melting procedures for grey cast iron	**07**
		OR	
8.	**a.**	Discuss the pollution controls norms for foundry industries	**06**
	b.	Discuss the need for modernization, and mechanization for moulding.	**08**